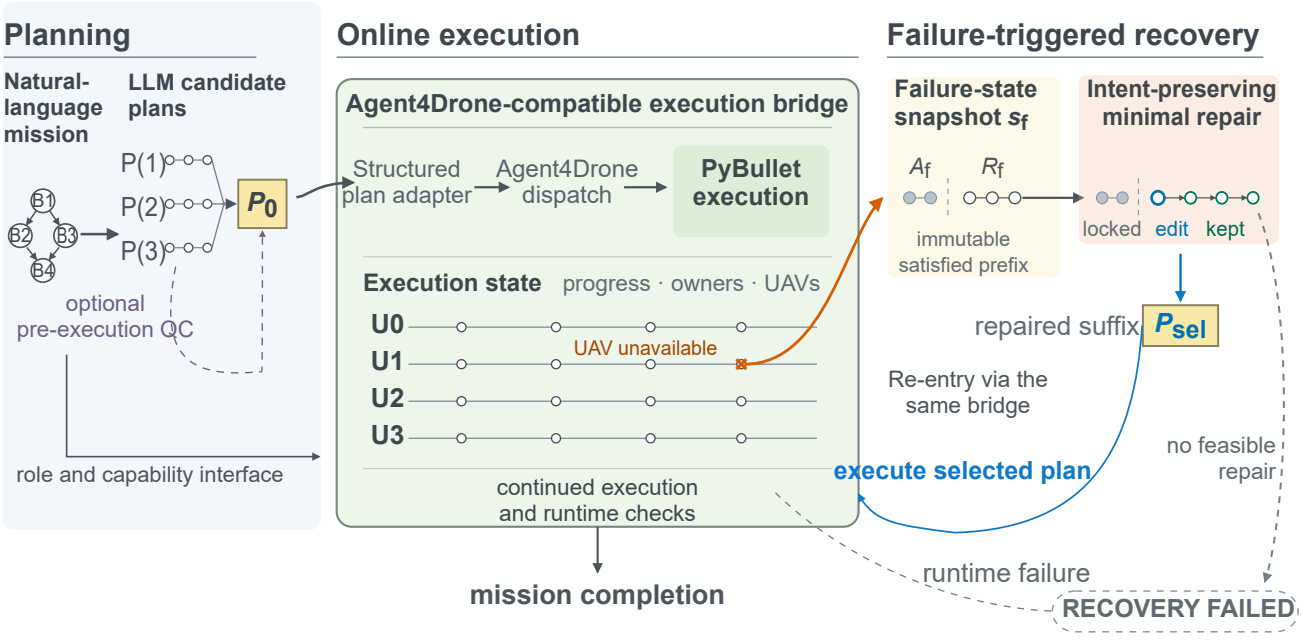

Planning
Natural-language mission
LLM candidate plans
B1
B2
B3
B4
P(1)
P(2)
P(3)
P0
optional pre-execution QC
role and capability interface
Online execution
Agent4Drone-compatible execution bridge
Structured plan adapter
Agent4Drone dispatch
PyBullet execution
Execution state
progress · owners · UAVs
U0
U1
U2
U3
UAV unavailable
continued execution and runtime checks
mission completion
Failure-triggered recovery
Failure-state snapshot $s_f$
$A_f$
$R_f$
immutable satisfied prefix
Intent-preserving minimal repair
locked
edit
kept
repaired suffix
$P_{sel}$
Re-entry via the same bridge
execute selected plan
no feasible repair
runtime failure
RECOVERY FAILED